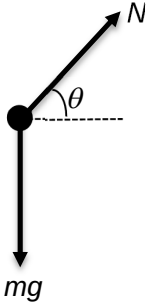


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|          | $= 7.77 \times 10^4 \text{ V}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [1] for correct answer                                                                                                            |
| 8(a)     | <p>The velocity changes as the direction of motion changes with time. So the motion is accelerated. By Newton's second law, a resultant force must be acting on it.</p> <p>Since there is no change in its speed, there is no component of resultant force tangential to the motion, resultant force must be perpendicular to its motion (i.e. towards the centre of the motion).</p> <p><i>Comments:</i><br/> <i>Many students did not explain why the force is pointing towards the centre.</i></p>                                                                                                                                                                                                                  | <p>[1]</p> <p>[1]</p>                                                                                                             |
| 8(b)(i)  | <p><math>r = 0.08 \cos(22^\circ) = 0.074175</math></p> <p><math>N \sin \theta = mg \dots \dots \dots (1)</math></p> <p><math>N \cos \theta = \frac{mv^2}{r} \dots \dots \dots (2)</math></p> <p>(1)/(2):</p> <p><math>\Rightarrow \tan \theta = \frac{gr}{v^2}</math></p> <p><math>\Rightarrow v = \sqrt{\frac{9.81 \times 0.074175}{\tan(22^\circ)}} = 1.3420 \approx 1.34 \text{ ms}^{-1}</math></p>  <p><i>Comments:</i><br/> <i>The weak students had problems setting up the two correct equations and calculating the radius of the circular motion. These students often did not draw free body diagram for analysis.</i></p> | <p>[1] for correct r</p> <p>[1] for the 2 correct equations</p> <p>[1] for correct substitution</p> <p>[1] for correct answer</p> |
| 8(b)(ii) | No as there is no vertical component of the normal contact force to balance the weight.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | [1] for correct answer                                                                                                            |
| 8(c)(i)  | <p>Gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point.</p> <p><i>Comments:</i><br/> <i>Common mistake includes writing just "work done" instead of "work done per unit mass".</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | [2]                                                                                                                               |

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| 8(c)(ii)       | $g = \frac{GM}{(R+h)^2} \Rightarrow GM = g(R+h)^2$ $\phi = -\frac{GM}{(R+h)} = -\frac{g(R+h)^2}{(R+h)} = -g(R+h)$                                                                                                                                                                                                                                                                                                   | <p>[1] for correct expression</p> <p>[1] for correct manipulation</p>                                          |
| 8(c)(iii)      | $\phi = -g(R+h) = -0.15(3.3 \times 10^6 + 1.4 \times 10^7) = -2595000$ <p>Gain in gravitational potential energy = Loss in kinetic energy</p> $m(2595000) = \frac{1}{2}mv^2$ $\Rightarrow v = 2278.2 \approx 2280 \text{ m s}^{-1}$ <p><i>Comments:</i><br/>Some students thought that the space probe was moving in circular motion and hence started using the wrong equations (e.g. <math>F_g = F_c</math>).</p> | <p>[1] for correct potential at P</p> <p>[1] for correct method/substitution</p> <p>[1] for correct answer</p> |
| 8(c)(iv)<br>1. | $\Delta U = m(\Delta\phi) = 12000 \times [(-3.2) - (-12.6)] \times 10^6 \approx 1.13 \times 10^{11} \text{ J}$ <p><i>Comments:</i><br/>Common mistakes include calculating the potential energy at <math>4R</math> instead of change of potential energy; calculating just the potential; using <math>\text{J kg}^{-1}</math> instead of <math>\text{MJ kg}^{-1}</math>.</p>                                        | <p>[1] for correct substitution</p> <p>[1] for correct answer</p>                                              |
| 8(c)(iv)<br>2. | $g = -\frac{d\phi}{dr} = -\frac{\Delta\phi}{\Delta r} = -\frac{(-1.0) - (-5.2)}{6.5 - 1.5} \times \frac{10^6}{3.3 \times 10^6} \approx 0.255 \text{ N kg}^{-1}$ <p><i>Comments:</i><br/>Common mistakes include calculating the gradient without taking note of the units on the axis; forgetting to substitute the value of <math>R</math>.</p>                                                                    | <p>[1] for correct substitution</p> <p>[1] for correct answer</p>                                              |
| 8(c)(iv)<br>3. | $g = \frac{GM}{(4R)^2} = \frac{GM}{16R^2} = 0.255 \text{ N kg}^{-1}$ $g_{\text{surface}} = \frac{GM}{(R)^2} = 0.255 \times 16 = 4.08 \text{ N kg}^{-1}$ <p><i>Comments:</i><br/>Most students did not understand that the gravitational field strength increases when distance decreases and <math>g \propto \frac{1}{r^2}</math>.</p>                                                                              | <p>[1] for correct method</p> <p>[1] for correct answer</p>                                                    |
| 8(a)           | The volt is the potential difference between two points in a circuit in                                                                                                                                                                                                                                                                                                                                             | [1] for correct                                                                                                |